

Machine Technical Manual: High-Precision Wafer Dicing Saw

Domain Intelligence & Anomaly Diagnostics Specification for Model DIC-001

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SYNTHETIC PROTOTYPE TECHNICAL MANUAL — VECTORAI SYSTEM SPECIFICATION

This document is artificially generated specifically for the VectorAI predictive maintenance demonstration, knowledge base retrieval (RAG), formula-based RUL calculation, and automated diagnostics.

The specifications, thresholds, sensor values, service-life hours, maintenance intervals, and operating parameters are synthetic and must NOT be used for real-world physical industrial equipment operation or maintenance.

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1. Document Information & Governance

This technical document provides authoritative machine specifications for **High-Precision Wafer Dicing Saw** (Identifier: DIC-001, Manual ID: VAI-MAN-DIC-001). It establishes the single source of truth for the VectorAI platform, defining telemetry sensor schemas, threshold limits, maintenance intervals, and deterministic degradation parameters.

Governance Principle: All diagnostic algorithms, RUL models, and threshold engines in VectorAI must strictly adhere to the limits and parameters defined herein. No machine learning regressions or stochastic models are permitted in the primary RUL calculation pipeline.

2. Machine Overview & Manufacturing Process

Process Stage: Wafer Prep & Dicing

Precision dual-spindle wafer dicing system designed for cutting incoming 200mm/300mm semiconductor silicon, GaAs, and GaN wafers into individual micro-dies using high-speed diamond blade spindles operating up to 60,000 RPM with deionized coolant delivery.

Detailed Process Flow:

The dicing process utilizes resinoid/nickel-bond diamond blades mounted on high-frequency air-bearing spindles. Silicon wafers mounted on dicing tape are secured on a porous ceramic vacuum chuck table. High-pressure DI water jets cool the cutting zone, remove silicon swarf, and lubricate the blade kerf while optical cameras align scribe lanes with sub-micron precision.

Key Machine Subsystems:

- Air-Bearing High-Speed Spindle Assembly
- Porous Ceramic Vacuum Chuck & Work Stage
- DI Water Coolant Delivery & Jet Nozzle System
- High-Resolution Vision Alignment & Kerf Inspection Optical System
- Diamond Blade Flange & Auto-Contact Touch Sensor
- Dressing Board Mechanism for Blade Sharpening

3. Major Components & Degradation Characteristics

Component	Primary Function	Important Parameters	Degradation Indicators
Air-Bearing Spindle Assembly	Drives diamond cutting blade at rotational speeds from 30,000 to 60,000 RPM with minimal radial runout.	Spindle rotation speed (RPM), radial vibration (mm/s), motor winding current (A), bearing air pressure (MPa).	Increased high-frequency harmonic vibration (>0.8 mm/s), spindle bearing temperature rise, motor current spikes.
Diamond Dicing Blade	Precision mechanical shearing and material removal along scribe lanes on silicon and compound semiconductor wafers.	Blade outer diameter (mm), diamond grit exposure, kerf width (μm), blade wear index (μm).	Increased cutting resistance, wafer backside chipping (>25 μm), kerf width widening, blade edge loading.
Coolant Delivery System	Directs constant laminar DI water flow to blade kerf to dissipate cutting friction heat and flush silicon particulate.	Coolant temperature (°C), nozzle pressure (bar), volumetric flow rate (L/min), resistivity (MΩ·cm).	Nozzle pressure drop (<1.5 bar), coolant temperature rise (>28°C), nozzle orifice particulate fouling.
Porous Ceramic Vacuum Chuck	Immobilizes wafer dicing tape carrier securely during high-velocity cutting passes.	Holding vacuum level (kPa), chuck table flatness (μm), surface temperature (°C).	Vacuum leakage (>-60 kPa), micro-vibration transfer, particulate clogging in ceramic pores.
Vision Alignment Optical System	Performs dual-axis sub-pixel recognition of wafer alignment targets and post-cut kerf placement accuracy.	Illumination intensity (lux), focal distance (mm), pattern match confidence score (%).	Alignment search timeouts, optical lens dust contamination, LED strobe brightness decay.

4. Sensor Configuration & Telemetry Mapping

The following sensor schema represents the continuous telemetry parameters streamed by the machine controller into the VectorAI Edge Collector. Direction indicates whether increasing or decreasing values signify degradation.

Sensor ID	Sensor Name	Unit	Normal Range	Warning Range	Critical Range	Direction
vibration_spindle	Spindle Radial Vibration	mm/s	0.1 – 0.5	0.5 – 0.8	0.8 – 2.0	HIGHER IS WORSE
temperature_coolant	DI Water Coolant Temp	°C	18.0 – 24.0	24.0 – 28.0	28.0 – 50.0	HIGHER IS WORSE
load_motor	Spindle Motor Load	%	30.0 – 65.0	65.0 – 80.0	80.0 – 100.0	HIGHER IS WORSE
pressure_coolant	Coolant Nozzle Pressure	bar	2.2 – 3.5	1.5 – 2.2	0.0 – 1.5	LOWER IS WORSE
blade_wear_index	Diamond Blade Wear Index	µm	0.0 – 15.0	15.0 – 30.0	30.0 – 50.0	HIGHER IS WORSE
chuck_vacuum	Work Chuck Vacuum	kPa	-90.0 – -75.0	-75.0 – -60.0	-60.0 – 0.0	HIGHER IS WORSE

5. Sensor Threshold Definitions & Operational Boundaries

Sensor / Metric	Normal Operational Band	Warning Threshold Band	Critical Fault Band
Spindle Radial Vibration vibration_spindle	0.1 to 0.5 mm/s Nominal spindle dynamic balance and hydrodynamic air film.	0.5 to 0.8 mm/s Early bearing air film instability or minor blade unbalance.	0.8 to 2.0 mm/s Severe bearing wear or imminent air-bearing spindle crash.
DI Water Coolant Temp temperature_coolant	18.0 to 24.0 °C Optimal thermal stabilization for blade lubrication and wafer cooling.	24.0 to 28.0 °C Heat exchanger thermal drift or partial chiller degradation.	28.0 to 50.0 °C Severe thermal dissipation failure causing kerf thermal stress and micro-cracking.
Spindle Motor Load load_motor	30.0 to 65.0 % Standard cutting torque through silicon substrate.	65.0 to 80.0 % Diamond blade dulling or excessive feed rate resistance.	80.0 to 100.0 % Severe blade loading, spindle stalling risk, or blade breakage hazard.
Coolant Nozzle Pressure pressure_coolant	2.2 to 3.5 bar Adequate jet velocity to evacuate kerf swarf particles.	1.5 to 2.2 bar Filter restriction or partial nozzle orifice clogging.	0.0 to 1.5 bar Inadequate coolant supply causing immediate blade burning and wafer thermal shock.
Diamond Blade Wear Index blade_wear_index	0.0 to 15.0 µm Normal blade exposure and sharp diamond cutting facets.	15.0 to 30.0 µm Substantial matrix wear requiring dressing cycle or replacement schedule.	30.0 to 50.0 µm Blade end-of-life; risk of blade breakout, kerf drift, and die damage.
Work Chuck Vacuum chuck_vacuum	-90.0 to -75.0 kPa Firm vacuum clamp holding wafer tape flat during cutting stroke.	-75.0 to -60.0 kPa Minor seal leak or tape micro-lifting risk at wafer edges.	-60.0 to 0.0 kPa Insufficient vacuum clamping resulting in wafer shift or die flying hazard.

6. Standard Operating Conditions & Environmental Limits

Ambient Temperature:	20.0 - 24.0 °C (Cleanroom ISO Class 5 / Class 100)
Relative Humidity:	40% - 55% RH non-condensing
Normal Operating Temperature:	20.0 - 25.0 °C
Operating Pressure:	0.55 - 0.65 MPa (Clean Dry Air for Air Bearing)
Operating Speed / Velocity:	40,000 - 60,000 RPM
Standard Cycle Time:	45 - 75 seconds per 300mm wafer (standard die pitch)
Production Schedule:	20 - 24 hours/day continuous production
Recommended Operating Conditions:	Spindle warm-up cycle for 15 minutes before first cut; DI water resistivity > 15 MΩ·cm.
Max Continuous Operation:	168 hours (7 days) before scheduled dressing and optical calibration

7. Maintenance Schedule & Expected Service Life

Component	Interval (Hours)	Expected Life	Required Maintenance Action & Procedure
Diamond Dicing Blade	250 hrs	400 hrs	Inspect blade exposure, perform dressing board sharpening or install replacement blade with torque wrench calibration. <i>Procedure:</i> Use optical microscope to inspect blade edge. Dress blade using GC-1200 dressing plate for 5 passes at 20 mm/s.
Air-Bearing Spindle Flange	1000 hrs	6000 hrs	Check spindle axial and radial runout with capacitive displacement probe; clean flange mating surface. <i>Procedure:</i> Verify runout is < 0.5 µm. Inspect air bearing filtration unit and dew point.
Coolant Delivery & Nozzle Manifold	500 hrs	4000 hrs	Ultrasonic clean nozzle orifices, flush supply lines, and replace 0.2 µm DI water cartridge filter. <i>Procedure:</i> Remove nozzle bar, sonicate in 5% citric acid for 15 min, rinse with DI water, verify spray fan pattern.
Porous Ceramic Chuck Table	750 hrs	8000 hrs	Perform chemical back-flush and ultrasonic surface wipe down to eliminate embedded silicon slurry. <i>Procedure:</i> Apply isopropyl alcohol with lint-free cleanroom wipe, run back-purge vacuum cycle at +0.2 MPa.
Vision Optical Lens & Camera	350 hrs	10000 hrs	Clean front objective lens with optical solvent and execute sub-pixel reticle recalibration. <i>Procedure:</i> Use standard calibration grid target; verify telecentric distortion is < 0.05%.

8. Degradation Indicators & Physics of Failure

Each parameter below reflects physical mechanical or electrical wear in machine subsystems. The VectorAI anomaly engine monitors these indicators to calculate Remaining Useful Life (RUL).

Parameter	Normal Baseline	Degraded Condition	Critical Limit	Underlying Physics / Mechanism
Spindle Radial Vibration	Vibration amplitude between 0.10 and 0.50 mm/s with stable spectral baseline.	Vibration levels drift into 0.50 - 0.80 mm/s due to diamond blade micro-chipping or bearing air film instability.	Vibration exceeds 0.80 mm/s; mechanical imbalance causes severe wafer kerf damage.	Indicates physical loss of diamond matrix uniformity, blade flange clamping looseness, or bearing air clearance degradation.
Spindle Motor Load	Motor load steady between 30% and 65% during standard cutting feed.	Motor load climbs to 65% - 80% as cutting resistance increases.	Motor load exceeds 80%; spindle inverter experiences overcurrent and risks stall.	Indicates diamond grit dulling/loading, where friction replaces clean shearing, transferring excess mechanical resistance to spindle motor.
DI Water Coolant Temperature	Coolant temperature maintained strictly within 18.0 - 24.0 °C.	Temperature climbs to 24.0 - 28.0 °C as heat exchanger efficiency declines.	Coolant temperature exceeds 28.0 °C; silicon substrate suffers thermal expansion and stress micro-fractures.	Indicates fouling of heat exchanger or chiller refrigeration loop decay, degrading heat transfer away from blade cutting zone.
Coolant Nozzle Pressure	Continuous nozzle line pressure between 2.2 and 3.5 bar.	Pressure drops to 1.5 - 2.2 bar due to upstream filter loading.	Pressure drops below 1.5 bar; laminar cooling jet collapses into turbulent mist.	Indicates particulate clogging in line filters or nozzle aperture buildup, starving the cutting interface of lubrication.
Diamond Blade Wear Index	Radial blade wear below 15 µm.	Blade wear between 15 and 30 µm; kerf profile begins widening.	Blade wear exceeds 30 µm; blade diameter reduced to core substrate limit.	Represents the direct physical attrition of synthetic diamond abrasive matrix through cumulative silicon cutting distance.

9. Deterministic Formula-Based RUL Model Configuration

Base Useful Life (BUL): 6000 operating hours

Mathematical Formula: VectorAI computes RUL deterministically without neural networks or stochastic regression. For each parameter *i*:

- For HIGHER_IS_WORSE: $d_i = (currentValue - healthyLimit) / (criticalLimit - healthyLimit)$
- For LOWER_IS_WORSE: $d_i = (healthyLimit - currentValue) / (healthyLimit - criticalLimit)$
- Clamp: $d_i = \max(0.0, \min(1.0, d_i))$
- Overall Degradation: $D_{total} = \sum (w_i \times d_i)$ [where $\sum w_i = 1.00$]
- Remaining Useful Life: $RUL = \max(0, BaseUsefulLife \times (1.0 - D_{total}))$

Parameter	Sensor ID	Unit	Weight (w_i)	Healthy Limit	Critical Limit	Direction
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Spindle Vibration	vibration_spindle	mm/s	0.30	0.5	1.2	HIGHER IS WORSE
Spindle Motor Load	load_motor	%	0.25	65.0	90.0	HIGHER IS WORSE
Coolant Temperature	temperature_coolant	°C	0.15	24.0	35.0	HIGHER IS WORSE
Coolant Pressure	pressure_coolant	bar	0.15	2.5	1.2	LOWER IS WORSE
Blade Wear Index	blade_wear_index	µm	0.15	15.0	35.0	HIGHER IS WORSE
TOTAL WEIGHT SUM	-	-	1.00 (100%)	-	-	VERIFIED OK

10. Troubleshooting & Failure Symptoms Table

The following troubleshooting matrix provides rapid diagnostic mappings between observable anomalies and root causes.

Symptom & ID	Severity	Related Sensors	Possible Causes	Recommended Corrective Action
Spindle Radial Vibration Exceedance SYM-DIC-01	High	vibration_spindle, load_motor	- Diamond blade dynamic imbalance or flange clamping runout - Air-bearing air supply pressure fluctuation or moisture contamination - Spindle internal ceramic bearing race wear	Halt dicing cycle, inspect blade flange torque (0.8 Nm), verify CDA dew point (-40°C), perform dynamic balance calibration.
Elevated Coolant Supply Temperature SYM-DIC-02	Medium	temperature_coolant	- DI water chiller heat exchanger bio-fouling or sediment accumulation - Coolant circulation pump flow rate reduction - Chiller refrigerant leak or compressor duty overload	Inspect chiller loop temperature, clean heat exchanger plates, verify 4.5 L/min minimum flow rate.
Spindle Motor High Load Surge During Cut SYM-DIC-03	High	load_motor, blade_wear_index	- Diamond blade glazing and lost abrasive grit exposure - Work table feed velocity exceeding blade cutting capacity - Insufficient DI water lubrication at cutting kerf	Execute 5-pass dressing cycle on GC board, reduce feed rate by 15%, inspect coolant nozzle alignment.
Coolant Jet Pressure Drop Below Limit SYM-DIC-04	Critical	pressure_coolant, temperature_coolant	- Inline 0.2 µm DI water particulate filter clogged with silicon fines - Coolant delivery solenoid valve coil failure - Nozzle orifice blockage from crystallized mineral deposits	Immediately replace filter cartridge, perform ultrasonic purge of nozzle manifold, confirm 2.8 bar baseline pressure.
Work Chuck Vacuum Pressure Degradation SYM-DIC-05	Critical	chuck_vacuum	- Ceramic chuck surface micro-porosity contamination with slurry - Vacuum generator ejector venturi nozzle wear - Dicing tape puncture or bottom seal perimeter leak	Perform chemical back-flush of chuck with IPA, inspect ejector vacuum generator, verify wafer frame tape seal tension.

11. Comprehensive Diagnostic Knowledge Base (Failure Scenarios)

This section contains 13 detailed synthetic failure scenarios used as the initial primary diagnostic knowledge base by the VectorAI Machine Agent. When an anomaly is detected, the agent queries this manual first before fallback to RAG.

Scenario 1: High-Frequency Harmonic Spindle Vibration (SCEN-DIC-001)	Severity: HIGH
Telemetry Sensor Pattern:	vibration_spindle rises from 0.42 to 0.88 mm/s; load_motor stable at 48%
Possible Root Causes:	Diamond blade edge micro-chipping Spindle flange tightening screw relaxation Spindle shaft air film thickness reduction
Recommended Corrective Action:	Retorque blade clamping flange to 0.8 Nm with calibrated torque wrench. If vibration persists, replace blade.
Verification & Recovery Steps:	Run 10,000 RPM idle spin test > Measure radial runout with capacitive sensor > Execute dry run without wafer load
Scenario 2: Diamond Blade Loading & Cutting Resistance Spike (SCEN-DIC-002)	Severity: HIGH
Telemetry Sensor Pattern:	load_motor surges from 52% to 84%; blade_wear_index jumps to 28 µm
Possible Root Causes:	Silicon swarf embedded in diamond matrix resinoid pores Excessive feed velocity on thick wafer substrate Improper coolant spray angle
Recommended Corrective Action:	Initiate automated blade dressing sequence on silicon carbide block for 8 strokes. Adjust nozzle angle to 45°.
Verification & Recovery Steps:	Verify motor load returns to < 55% during test cut > Inspect kerf edge for burrs under 50x magnification
Scenario 3: Coolant Delivery Manifold Pressure Drop (SCEN-DIC-003)	Severity: CRITICAL
Telemetry Sensor Pattern:	pressure_coolant drops from 2.8 bar to 1.3 bar; temperature_coolant rises by 4.2°C
Possible Root Causes:	Main 0.2 µm DI filter differential pressure threshold exceeded Coolant pump impeller cavitation Nozzle feed line kink
Recommended Corrective Action:	Replace primary and secondary coolant filter cartridges. Purge trapped air from manifold.
Verification & Recovery Steps:	Check pressure gauge reads > 2.5 bar at full flow > Inspect spray symmetry across dual nozzles
Scenario 4: Wafer Micro-Shift During High-Speed X-Feed (SCEN-DIC-004)	Severity: CRITICAL
Telemetry Sensor Pattern:	chuck_vacuum degrades from -82 kPa to -54 kPa
Possible Root Causes:	Silicon slurry accumulation inside porous ceramic chuck channels Vacuum solenoid valve seal leak Improper wafer dicing tape lamination pressure
Recommended Corrective Action:	Perform hot DI water and IPA reverse purge of ceramic chuck table. Test vacuum holding force.
Verification & Recovery Steps:	Verify vacuum level drops to < -80 kPa with test frame > Perform pull force gauge test (> 15 N)

Scenario 5: Spindle Thermal Expansion and Z-Axis Kerf Depth Drift (SCEN-DIC-005)	Severity: MEDIUM
Telemetry Sensor Pattern:	vibration_spindle at 0.65 mm/s; load_motor at 72%; coolant temp at 27.5°C
Possible Root Causes:	Spindle internal chiller coolant jacket restriction Continuous operation exceeding 12 hours without thermal stabilization Room cleanroom ambient temperature fluctuation
Recommended Corrective Action:	Recalibrate non-contact height sensor baseline. Clean spindle cooling jacket circuit.
Verification & Recovery Steps:	Perform touch-sensor zero calibration > Check kerf depth on test wafer (nominal $\pm 2 \mu\text{m}$)
Scenario 6: Excessive Wafer Backside Chipping on Die Corners (SCEN-DIC-006)	Severity: HIGH
Telemetry Sensor Pattern:	blade_wear_index at 32 μm ; vibration_spindle at 0.74 mm/s
Possible Root Causes:	Diamond blade wear past nominal diamond matrix boundary Blade kerf tapering due to side-face erosion Dicing tape adhesion loss from coolant over-penetration
Recommended Corrective Action:	Replace diamond blade immediately. Inspect wafer backing tape UV cure parameters.
Verification & Recovery Steps:	Measure backside chipping width on sample die ($< 15 \mu\text{m}$ allowable) > Confirm new blade zero reference
Scenario 7: Spindle Inverter Harmonic Overcurrent Trip (SCEN-DIC-007)	Severity: CRITICAL
Telemetry Sensor Pattern:	load_motor spikes to 96% instantaneously; vibration_spindle at 1.1 mm/s
Possible Root Causes:	Blade jamming caused by silicon fragment wedged in blade cover Spindle inverter IGBT phase imbalance Spindle bearing air pressure dropped below interlock limit
Recommended Corrective Action:	Emergency stop. Clear blade chamber of fragments. Inspect air supply regulator (min 0.55 MPa).
Verification & Recovery Steps:	Manually spin spindle by hand with air on (must rotate freely) > Reset inverter faults > Perform slow ramp-up test
Scenario 8: Coolant Chiller Loop Heat Saturation (SCEN-DIC-008)	Severity: HIGH
Telemetry Sensor Pattern:	temperature_coolant drifts continuously up to 31.2°C; pressure_coolant stable
Possible Root Causes:	External facility chilled water loop temperature elevated Chiller condenser coil dust clogging Refrigerant R134a undercharge
Recommended Corrective Action:	Notify facilities team to check primary loop. Clean chiller condenser filters. Verify temperature returns to 21°C.
Verification & Recovery Steps:	Monitor temperature log over 30 min > Verify stability at $22.0 \pm 0.5^\circ\text{C}$
Scenario 9: Vision Alignment Optical Target Misrecognition (SCEN-DIC-009)	Severity: MEDIUM
Telemetry Sensor Pattern:	Cycle time increases by 25 seconds per wafer; sensor telemetry nominal
Possible Root Causes:	Microscope front protective glass coated with dried silicon aerosol mist Ring light LED intensity degradation Wafer surface reflectivity change from new metal layer
Recommended Corrective Action:	Clean microscope protective glass with optical tissue and lens cleaner. Run illumination auto-calibration.
Verification & Recovery Steps:	Run vision recognition test on standard wafer pattern > Confirm match score > 95%

Scenario 10: Work Table Y-Axis Ball Screw Linear Pitch Error (SCEN-DIC-010)	Severity: HIGH
Telemetry Sensor Pattern:	Kerf positioning offset increases linearly across wafer diameter (+8 µm)
Possible Root Causes:	Y-axis precision linear guide lubrication starvation Linear optical scale glass grating contamination Ball screw pre-load spring fatigue
Recommended Corrective Action:	Apply cleanroom grease (Kluber Isoflex NBU 15) to linear guides. Clean optical scale reading head.
Verification & Recovery Steps:	Execute laser interferometer positioning verification > Verify repeatability < 1.0 µm
Scenario 11: Blade Dressing Board Auto-Mechanism Jam (SCEN-DIC-011)	Severity: MEDIUM
Telemetry Sensor Pattern:	load_motor at 68% during dressing cycle; dressing cycle timeout alert
Possible Root Causes:	Dressing board mechanical feed stage lead screw bound by slurry Dressing board material thickness exhausted Pneumatic actuator cylinder seal sticking
Recommended Corrective Action:	Replace dressing board plate. Clean and lubricate dressing stage pneumatic cylinder.
Verification & Recovery Steps:	Run manual dress test > Confirm touch sensor triggers at correct Z coordinate
Scenario 12: Air-Bearing Clean Dry Air Dew Point Alert (SCEN-DIC-012)	Severity: CRITICAL
Telemetry Sensor Pattern:	vibration_spindle exhibits micro-spikes (0.35 to 0.65 mm/s intermittently)
Possible Root Causes:	Compressed air desiccant dryer saturation Air filter oil coalescing element breakthrough Main air line condensate buildup
Recommended Corrective Action:	Immediately check air supply dew point sensor. Replace air filter sub-assembly before moisture ruins air bearings.
Verification & Recovery Steps:	Verify CDA dew point is < -40°C > Verify zero oil aerosol with particle counter
Scenario 13: Blade Touchdown Sensor Sensitivity Drift (SCEN-DIC-013)	Severity: MEDIUM
Telemetry Sensor Pattern:	Blade wear index calculation fluctuates erratically (± 10 µm between cuts)
Possible Root Causes:	Electrical grounding contact resistance increase at workpiece chuck Non-contact electrical discharge sensor probe oxidized Substrate tape dielectric thickness variation
Recommended Corrective Action:	Clean touchdown contact plate with scotch-brite pad and IPA. Calibrate electrical sensor threshold.
Verification & Recovery Steps:	Execute 5 repeated touchdown tests > Verify standard deviation < 1.0 µm

12. Standard Operating & Maintenance Procedures (SOP)

SOP-01: Shift Start Operational Inspection & Tool Touch-Off

1. Verify Clean Dry Air (CDA) supply pressure is within 0.55 - 0.65 MPa and dew point is < -40°C.
2. Ensure cleanroom ambient temperature is stabilized within specified operating window (20 - 24°C).
3. Run automated sensor zero-calibration and optical target homing sequence.
4. Process 2 calibration test pieces; inspect output dimensions and telemetry stability on VectorAI Dashboard.

SOP-02: Tool Replacement & Calibration Protocol

1. Lock out and tag out (LOTO) machine controller before entering mechanical enclosure.
2. Use calibrated preset torque wrench to fasten tool fasteners to exact manufacturer specification.
3. Execute optical non-contact tool alignment and measure runout / contact resistance.
4. Log maintenance record in VectorAI with work order ID, technician ID, and timestamp.

SOP-03: Anomaly Escalation Protocol

When a sensor crosses from Warning into Critical threshold band:

1. VectorAI generates automated Severity: High/Critical Incident.
2. System automatically references Section 11 Diagnostic Knowledge for recommended immediate actions.
3. If Remaining Useful Life (RUL) falls below 48 hours, dynamic production rerouting is triggered to alternate healthy lines.

13. Document Revision History

Revision	Release Date	Author / Organization	Description of Changes
v1.0	2026-08-26	VectorAI Knowledge Engineering	Initial synthetic technical specification and diagnostic manual release.
v0.9-RC	2026-08-15	VectorAI Systems Lab	Draft sensor thresholds, RUL weight configuration, and failure scenarios.

14. VectorAI RAG Integration & Knowledge Indexing

This document is formatted with distinct semantic headings, tables, and structured diagnostic scenarios to facilitate high-precision chunking and vector embedding in the VectorAI RAG pipeline.

- **Chunk Association:** Every indexed chunk retains metadata: machineId, manualId, section.
- **Scoped Retrieval:** Diagnostics for machine DIC-001 prioritize chunks where machineId == 'DIC-001'.